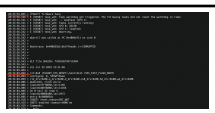
Recovery / Fault Injection Test Sheet

Webinar 5 — Sleep, WDT, Power Loss, Persistence

Environment

Field	Value
MCU	ESP32 DevKit V1
Sensor	DHT22 GPIO15
Button	GPIO4
LED	GPIO2
Relay	GPIO5
Serial	115200
Build	BUGGY

Test Matrix

Scenario	Risk	Expected	Actual	Evidence	P/F
Sleep 5	Timing / wake / persistence	Wake-up error margin ±15%; The device is operational after wake-up; Persistent data remains intact.	The time from [STATE] SLEEP to [BOOT] is ≈ 1.40 s (deviation ≈ −72%). Wake-up cause is correct and the device returns to the ACTIVE state. Persistent data remains intact.		F
WDT hang	Reset time / reason / recovery	The WDT reboots the device in approximately no more than 5 seconds. The reset reason makes it possible to identify a WDT reset. The device is operational after the reset. There is no boot loop.	WDT trigger ≈ 4.1 s. WDT full reset ≈ 11 s. The reset reason makes it possible to identify a WDT reset ([BOOT] reset_reason=INT_WDT). The device is operational after the reset. There is no boot loop.		WDT trigger ≈ 4.1 s → P WDT full reset ≈ 11 s → Not verified (simulator), hardware required. OBS-01
Relay + WDT	Safe actuator state	Actuator after WDT reset should be OFF	Actuator (relay) is ON after WDT		F

Interrupted save	Atomicity / integrity	After a reboot, either the old fully valid version or the new fully valid version of the state must remain. A mixed or partially updated state is not allowed. A corrupted state must be detected.	After reboot counter = 101 (expected 1 or a valid new value); other fields are unchanged. The corrupted state is not detected, no error is logged.		F
Sensor failure	Explicit invalid-data handling	The firmware must not treat this as a valid measurement; the logs must contain an explicit error; the actuator must not switch to an unsafe state because of invalid data.	The firmware treats temp=0.0°C and humidity=0.0% as a valid measurement, and there is no error in the logs.		F
Button	Single logical event	One physical button press must produce no more than one logical event.	Object non testable; The pins in diagram.json are specified as esp:4 instead of esp:D4		BLOCKED
Repeated recovery	No boot loop / stable	The device recovers correctly after each of the repeated resets (hang, reboot, powerloss). After every reset it returns to the ACTIVE state and responds to commands. No boot loop occurs, and the device remains stable. Heap does not decrease from cycle to cycle. Persistent data stays intact: after a reboot, either	Repeated recovery: 15 consecutive resets (hang ×5, reboot ×5, powerloss x5). After each one the device reached ACTIVE and responded to status. No boot loop observed; heap stayed = 336332. Note: the state integrity issue after powerloss during save is reported separately.	Video: https://drive.google.com/file/d/1GXpGsLyX_Ws6S2GuZfc_RwXW_fhcs_F6/view?usp=sharing	P

		the old fully valid state or the new fully valid state remains, and a corrupted state is detected.			

Bug Report

Bug-01

Title: Sleep 5 wakes the device after ≈1.4 s instead of 5 s ±15%

Requirement: R4

Risk: Timing / wake

Preconditions: Open the project in VS Code and launch the Wokwi simulator

Steps:

1. Type ‘sleep 5’ command
2. Measure the time from [STATE] SLEEP to [BOOT]

Expected: Wake-up error margin ±15% (R4 — Sleep / Wake)

Actual: The time from [STATE] SLEEP to [BOOT] is ≈ 1.40 s (deviation ≈ −72%).

Timing / Reset Reason: Timing: [STATE] SLEEP -> [BOOT] ≈ 1.40 s. Reset Reason: [BOOT] reset_reason=DEEP_SLEEP (rst:0x5 (DEEPSLEEP_RESET))

Evidence:

```
20:21:14.874 > [STATE] SLEEP timer=5 s
20:21:15.403 > ets Jul 29 2019 12:21:46
20:21:15.403 >
20:21:15.411 > rst:0x5 (DEEPSLEEP_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
20:21:15.412 > configsip: 0, SPIWP:0xee
20:21:15.412 > clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
20:21:15.424 > mode:DIO, clock div:2
20:21:15.424 > load:0x3fff0030,len:1156
20:21:15.425 > load:0x40078000,len:11456
20:21:15.425 > ho 0 tail 12 room 4
20:21:15.425 > load:0x40080400,len:2972
20:21:15.425 > entry 0x400805dc
20:21:16.278 > [BOOT] reset_reason=DEEP_SLEEP
20:21:16.278 > [WAKE] cause=4
20:21:16.278 > [WDT] enabled timeout=4000 ms
```

Severity: Major

Reproducibility: 5/5 Reproducible

Bug-02

Title: Actuator is ON after WDT reset

Requirement: R5

Risk: Safe actuator state

Preconditions: Open the project in VS Code and launch the Wokwi simulator

Steps:

1. Type ‘relay on’ command
2. Type ‘status’ command
3. Type ‘hang’ command
4. Type ‘status’ command
5. Compare the relay state before and after the WDT reset.

Expected: Actuator after WDT reset should be OFF

Actual: Actuator (relay) is ON after WDT

Timing / Reset Reason: Reset Reason: [BOOT] reset_reason=INT_WDT (rst:0x8 (TG1WDT_SYS_RESET))

Evidence:

```
  reboot
{"state":"ACTIVE","reset_reason":"DEEP_SLEEP","relay":true,"counter":4,"heap":336332,"sensor_ok":false}
{"state":"ACTIVE","reset_reason":"DEEP_SLEEP","relay":true,"counter":4,"heap":336332,"sensor_ok":false}
[CLI] ERROR unknown command
[SENSOR] temp=0.0C humidity=0.0%
                              [FAULT] firmware hang
E (63507) task_wdt: Task watchdog got triggered. The following tasks did not reset the watchdog in time:
E (63507) task_wdt:  - loopTask (CPU 1)
E (63507) task_wdt: Tasks currently running:
E (63507) task_wdt: CPU 0: IDLE0
E (63507) task_wdt: CPU 1: loopTask
E (63507) task_wdt: Aborting.

abort() was called at PC 0x400dbf11 on core 0


Backtrace: 0x400835b5:0x3ffbea8c |<-CORRUPTED




ELF file SHA256: 7428136fd4fc6204

ets Jul 29 2019 12:21:46

rst:0x8 (TG1WDT_SYS_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
entry 0x400805dc
[BOOT] reset_reason=INT_WDT
                           [WDT] enabled timeout=4000 ms
                                                    Commands:
  help
  status
  sensor
  relay on
  relay off
  save
  sleep <seconds>
  hang
  powerloss
  reboot
{"state":"ACTIVE","reset_reason":"INT_WDT","relay":true,"counter":4,"heap":336332,"sensor_ok":false}
```

Severity: Critical

Reproducibility: 5/5 Reproducible

Bug-03

Title: Corrupted state after power loss during save is not detected

Requirement: R6

Risk: Atomicity / integrity

Preconditions: Open the project in VS Code and launch the Wokwi simulator

Steps:

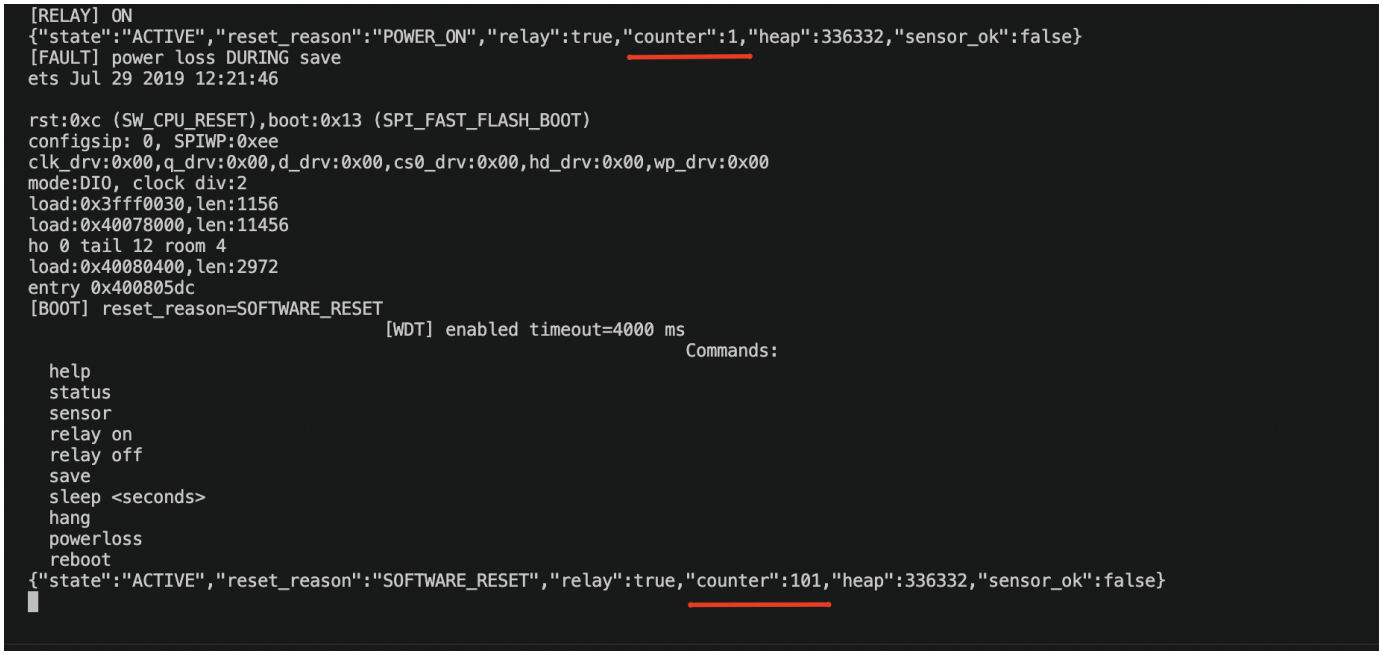
1. Type ‘relay on’ command
2. Type ‘status’
3. Type ‘powerloss’ and wait for reboot
4. Type ‘status’ and compare counter and relay with the values from step 2.

Expected: After a reboot, either the old fully valid version or the new fully valid version of the state must remain.A mixed or partially updated state is not allowed. A corrupted state must be detected.

Actual: After typing relay on, status shows counter=1, relay=true. After typing powerloss (log: [FAULT] power loss DURING save; the command simulates a reboot at the critical moment of the write) and waiting for the reboot, status shows counter=101, relay=true. The state is neither the old valid version (counter=1) nor a valid new value: the counter increased by 100 and holds a value that belongs to no valid version, while the other fields remain unchanged. This is a mixed (partially written) state. The firmware does not detect it and does not log any error. The reason for the +100 offset is not visible in the logs.

Timing / Reset Reason: Reset Reason: [BOOT] reset_reason=SOFTWARE_RESET (rst:0xc (SW_CPU_RESET))

Evidence:



Severity: Major

Reproducibility: 5/5 Reproducible

Bug-04

Title: Sensor returns temp=0.0°C and humidity=0.0% and the firmware accepts it as a valid measurement, no error logged

Requirement: R2

Risk: Explicit invalid-data handling

Preconditions: Open the project in VS Code and launch the Wokwi simulator

Steps:

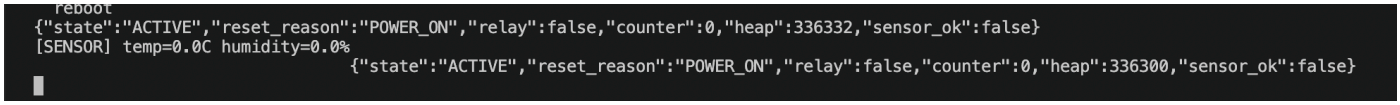
1. Type ‘sensor’ command
2. Type ‘status’ command
3. Pay attention to logs

Expected: The firmware must not treat this as a valid measurement; the logs must contain an explicit error; the actuator must not switch to an unsafe state because of invalid data.

Actual: The firmware treats temp=0.0°C and humidity=0.0% as a valid measurement, and there is no error in the logs.

Timing / Reset Reason: no reset

Evidence:



Severity: Major

Reproducibility: 5/5 Reproducible

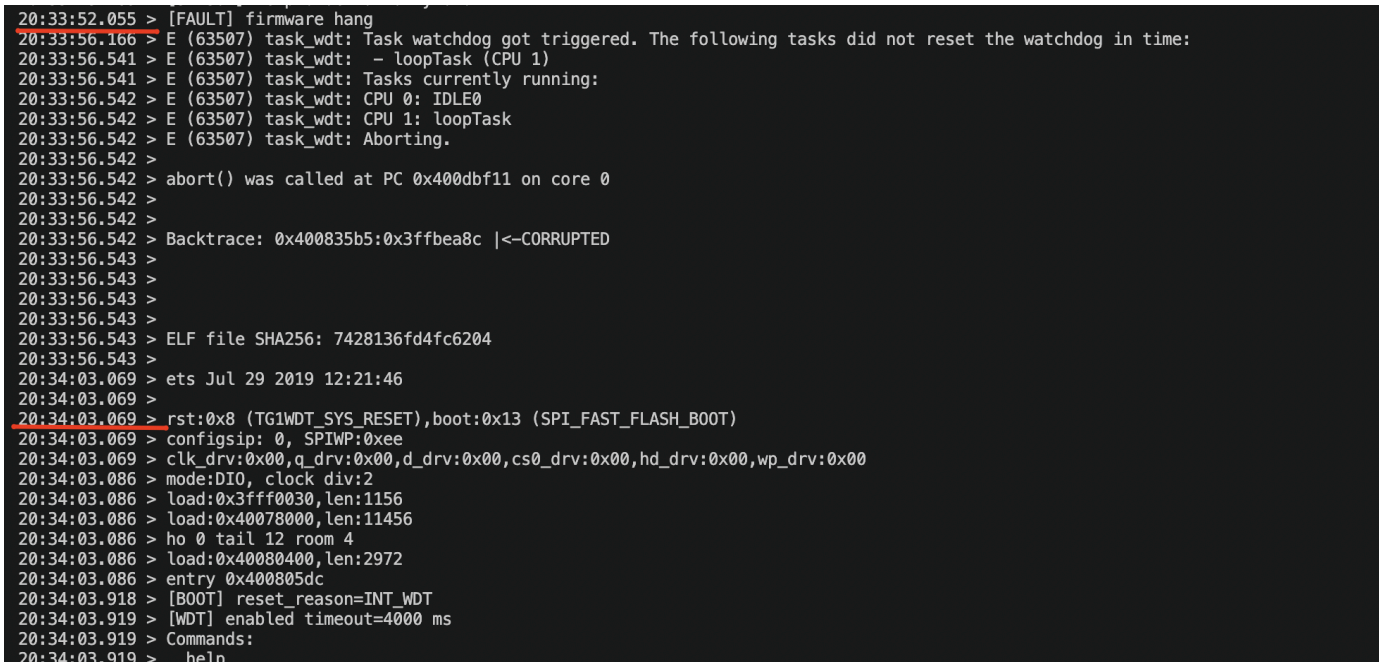
Observation

OBS-01 (needs hardware verification)

Title: WDT trigger is on time, but the full reset takes ≈ 11 s in the simulator

Description: After hang, the task watchdog triggers ≈ 4.1 s after [FAULT] firmware hang (WDT timeout = 4000 ms), which meets R5. The reset (rst:0x8, reset_reason=INT_WDT) appears ≈ 11.0 s after the fault; the extra ≈ 6.5 s falls between abort() and the next boot (panic handling and reboot in Wokwi). Timestamps are assigned by the host, not by the device, so the full reset time cannot be confirmed as a firmware defect.

Recommendation: Re-measure the full reset time on real hardware (e.g., with a logic analyzer or a GPIO toggle at fault and at boot).



OBS-02 (needs hardware verification)
Title: Extra rst:0x8 (TG1WDT_SYS_RESET) ≈ 0.65 s after sleep 30

Description: After [STATE] SLEEP timer=30 s, an unexpected WDT reset (reset_reason=INT_WDT) occurs ≈ 0.65 s later, before the deep-sleep reset. This may be a Wokwi artifact (deep-sleep entry vs. WDT handling in simulation) and was not confirmed on hardware.

Recommendation: Repeat sleep 30 on real hardware and check whether the extra reset occurs.

```
20:06:41.529 > [STORE] saved
20:06:41.530 > [STATE] SLEEP timer=30 s
20:06:42.183 > ets Jul 29 2019 12:21:46
20:06:42.183 >
20:06:42.183 > rst:0x8 (TG1WDT_SYS_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
20:06:42.187 > configsip: 0, SPIWP:0xee
20:06:42.187 > clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
20:06:42.194 > mode:DIO, clock div:2
20:06:42.195 > load:0x3fff0030,len:1156
20:06:42.195 > load:0x40078000,len:11456
20:06:42.195 > ho 0 tail 12 room 4
20:06:42.204 > load:0x40080400,len:2972
20:06:42.204 > entry 0x400805dc
20:06:43.072 > [BOOT] reset_reason=INT_WDT
20:06:43.075 > [WDT] enabled timeout=4000 ms
20:06:43.075 > Commands:
20:06:43.075 >   help
20:06:43.075 >   status
20:06:43.075 >   sensor
20:06:43.076 >   relay on
20:06:43.080 >   relay off
20:06:43.081 >   save
20:06:43.081 >   sleep <seconds>
20:06:43.081 >   hang
20:06:43.081 >   powerloss
20:06:43.081 >   reboot
20:06:44.615 > ets Jul 29 2019 12:21:46
20:06:44.615 >
20:06:44.615 > rst:0x5 (DEEPSLEEP_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
20:06:44.619 > configsip: 0, SPIWP:0xee
20:06:44.627 > clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
20:06:44.637 > mode:DIO, clock div:2
20:06:44.642 > load:0x3fff0030,len:1156
20:06:44.643 > load:0x40078000,len:11456
20:06:44.651 > ho 0 tail 12 room 4
20:06:44.652 > load:0x40080400,len:2972
20:06:44.663 > entry 0x400805dc
20:06:45.515 > [BOOT] reset_reason=DEEP_SLEEP
20:06:45.518 > [WAKE] cause=4
20:06:45.518 > [WDT] enabled timeout=4000 ms
20:06:45.518 > Commands:
20:06:45.518 >   help
20:06:45.518 >   status
20:06:45.518 >   sensor
20:06:45.532 >   relay on
20:06:45.533 >   relay off
20:06:45.533 >   save
20:06:45.533 >   sleep <seconds>
20:06:45.533 >   hang
20:06:45.533 >   powerloss
20:06:45.533 >   reboot
20:06:58.595 > {"state":"ACTIVE","reset_reason":"DEEP_SLEEP","relay":true,"counter":2,"heap":336332,"sensor_ok":false}
```

Release Decision

Decision	**NO-GO:** 1. sleep 5 wakes the device after ≈ 1.40 s (deviation ≈ −72%) (Bug-01). 2. Actuator (relay) is ON after WDT reset (required OFF) (Bug-02). 3. After a power loss during save, counter = 101 (expected 1 or a valid new value). The mixed state is not detected, and no error is logged (Bug-03). 4. The firmware treats temp=0.0°C and humidity=0.0% as a valid measurement, and there is no error in the logs (Bug-04).
Blocking risks	Safe actuator state: the relay remains ON after WDT reset.

	Atomicity / integrity: counter = 101 matches no valid version. The torn state is not detected, and no error is logged. Explicit invalid-data handling: 0.0°C / 0.0% is accepted as valid, and no error is logged.
Tests required on real hardwared	Sleep 5/30, including whether the extra WDT reset occurs after sleep 30, Powerloss, WDT reset, Repeated recovery. Peripherals: DHT22, button, LED and relay behavior (signal wires were not connected in the simulation).
Not confirmed in the simulator (hardware verification required)	Full WDT reset time ≈ 11 s (the WDT trigger itself occurs ≈ 4.1 s after the fault, which meets R5) (OBS-01). Extra rst:0x8 (TG1WDT_SYS_RESET) ≈ 0.65 s after sleep 30 (OBS-02).